

Visual Studio Code & Git Integration Guide

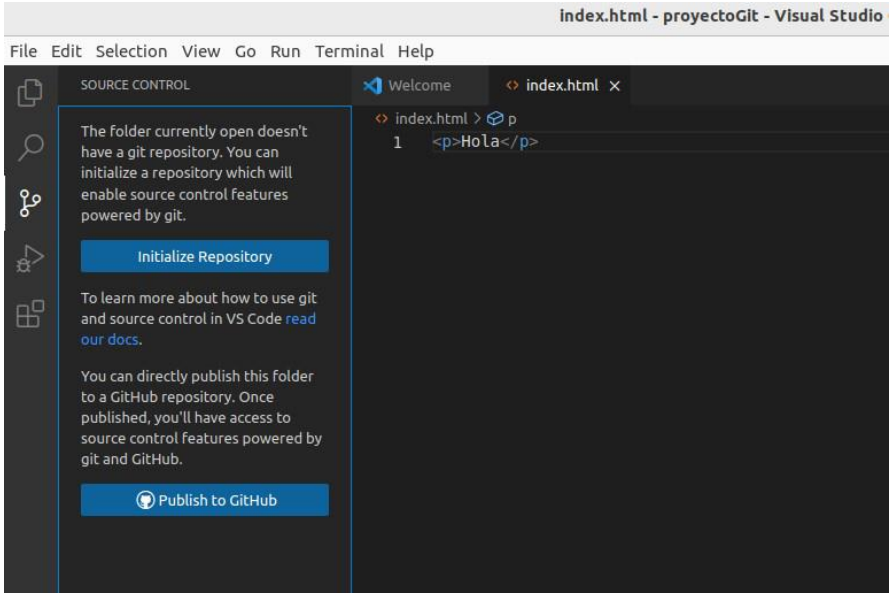
By Raúl Palao

Initialize a Local Repository

To create a local repository within your project, open Visual Studio Code and navigate to the Git sidebar. Select the 'Initialize Repository' option in the Source Control view. This action automatically generates a hidden .git folder in your project directory, enabling source control features.

Equivalent Terminal Command:

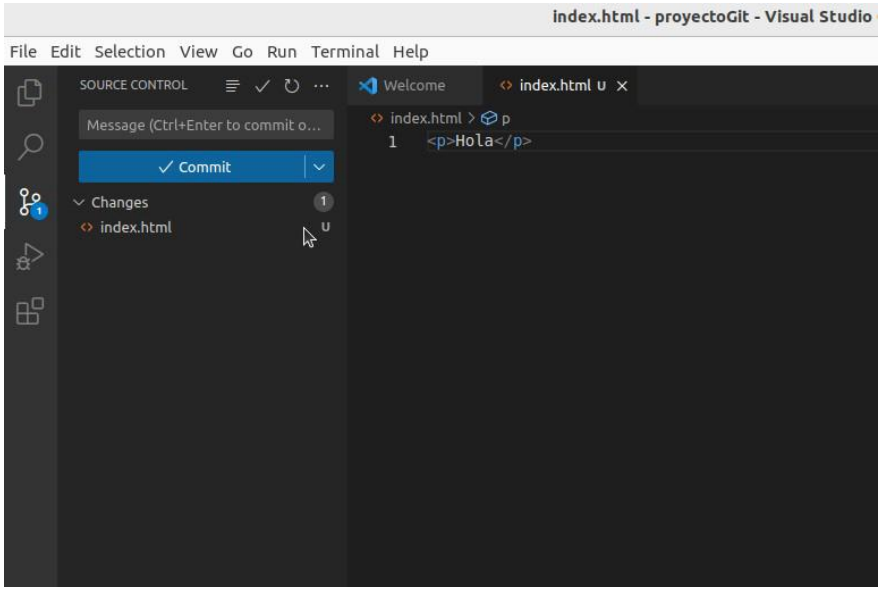
- `git init`



Staging Changes

Once your repository is initialized, you can begin making changes to your code, such as editing an index.html file. To stage these changes, navigate to the Source Control view. Next to each modified file, you will see a '+' icon. Click this '+' to add the specific files you want to the staging area.

```
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Equivalent Terminal Command:
git add index.html
```



Executing a Commit

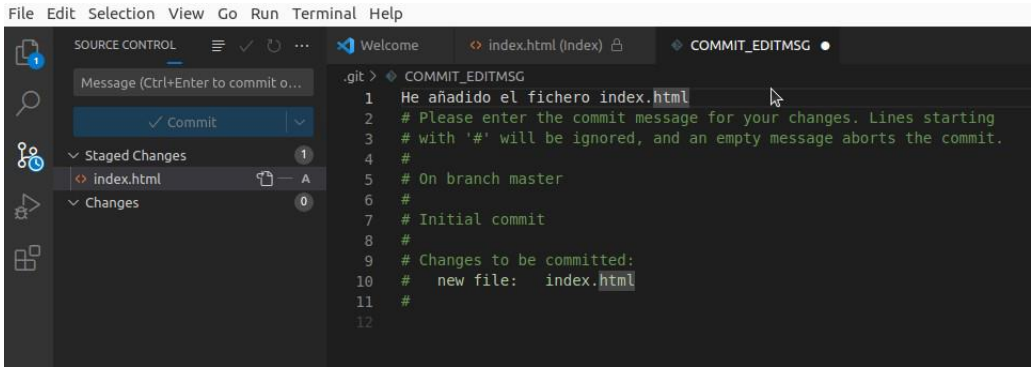
To execute a commit, click the Commit button in the Source Control view. This opens a new COMMIT_EDITMSG file containing information about the commit.

Write your commit message (e.g., "He añadido el fichero index.html") on the first line.

Saving and closing this file completes the commit process.

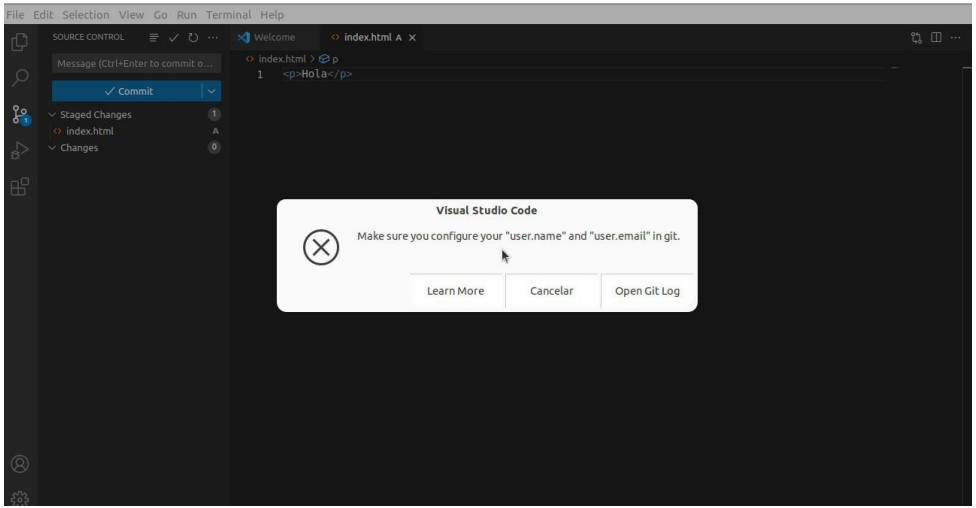
Equivalent Terminal Command

```
git commit -m "Añado el fichero index.html"
```



First Commit Identity Warning

If you attempt to make your first commit without configuring your Git identity, Visual Studio Code will display a warning message.



Mandatory Step: **This configuration is a mandatory step before any commits can be successfully recorded.**

Configuring Global Identity in VS Code

You can resolve the identity prompt using the integrated terminal in Visual Studio Code, which allows you to execute any Git command directly within the editor.

Open the terminal and configure your global repository name and email.

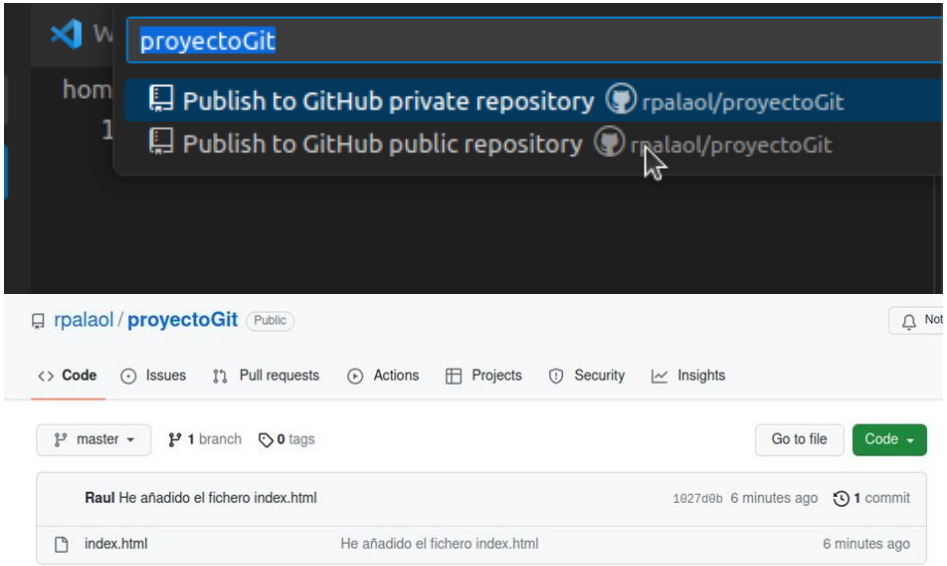
bash — VS Code Integrated Terminal

```
# Equivalent Terminal Commands:
$ git config --global user.name "Raul"
$ git config --global user.email "r.palaolozano@edu.gva.es"
```

Publishing to GitHub

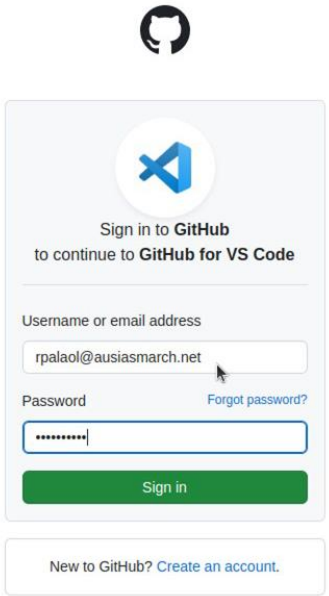
After authenticating, you will be presented with publishing options. Click on 'Publish to GitHub public repository'. Visual Studio Code will automatically create the remote repository on GitHub and upload your initial commit without requiring you to leave the editor.

```
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Equivalent Terminal Command:
  ● git push origin master
```



Authenticating Your Account

When you click the button to publish your repository for the first time, Visual Studio Code will require you to authenticate your account.



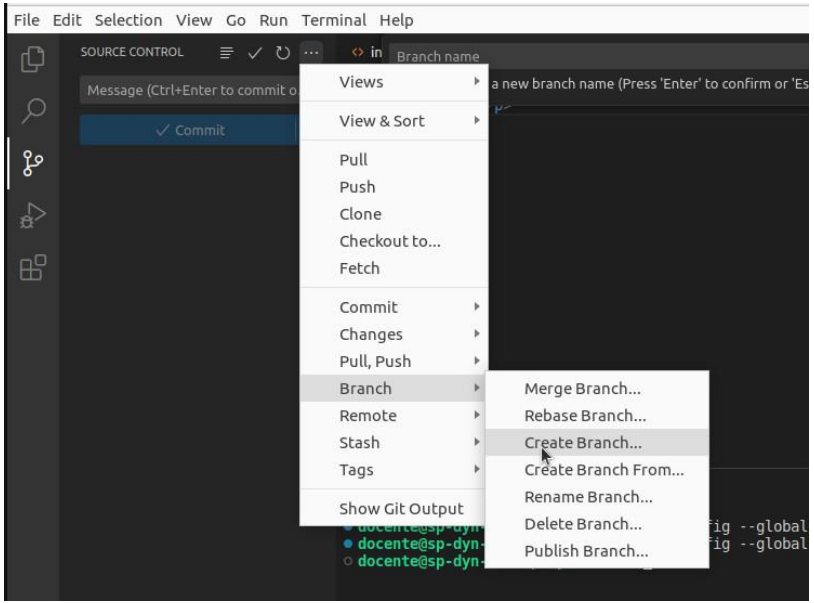
Creating & Managing Branches

To manage branches, use the top menu (three dots) in the Source Control view, which provides access to most Git tasks.

Navigate to Branch and select Create Branch... You will be prompted to enter a new branch name, such as feature1.

Equivalent Terminal Commands:

- `git branch feature1`
- `git checkout feature1`

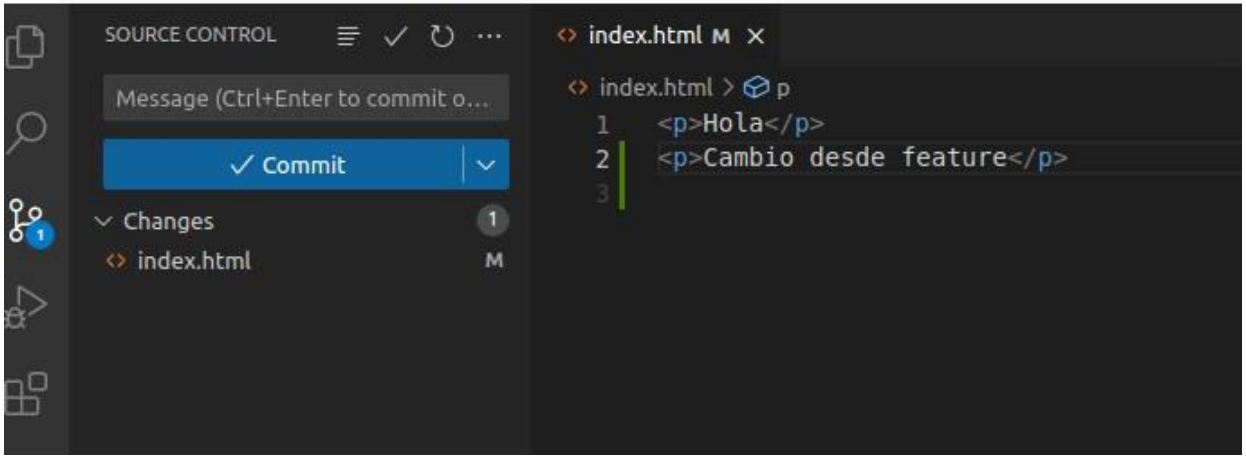


Modifying & Committing on a Feature Branch

Once your feature branch is created and active, you can continue modifying your project files. For example, adding a new line like:

```
<p>Cambio desde feature</p>
```

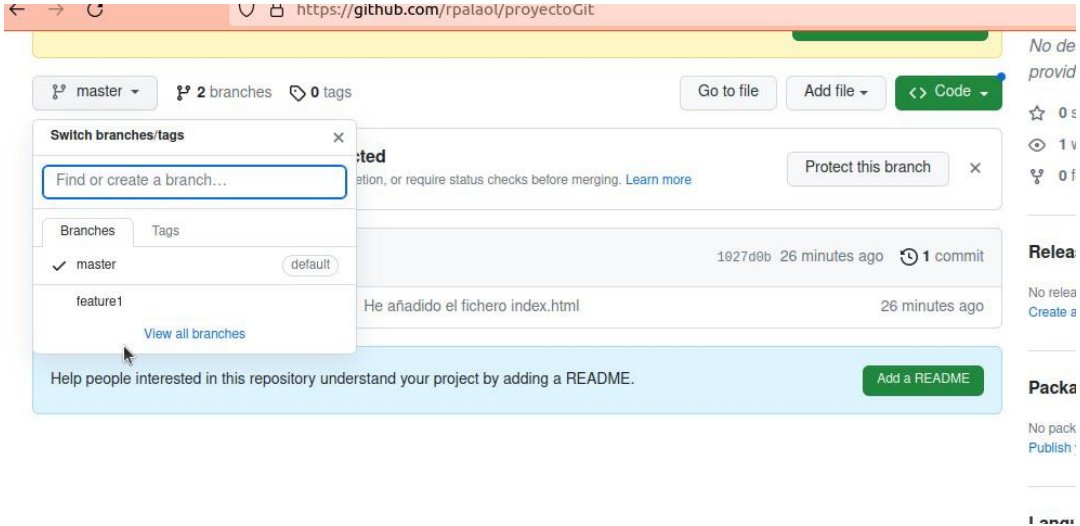
To stage and commit these new changes, simply repeat the previously described process of clicking the '+' icon and entering a commit message.



Publishing a Branch

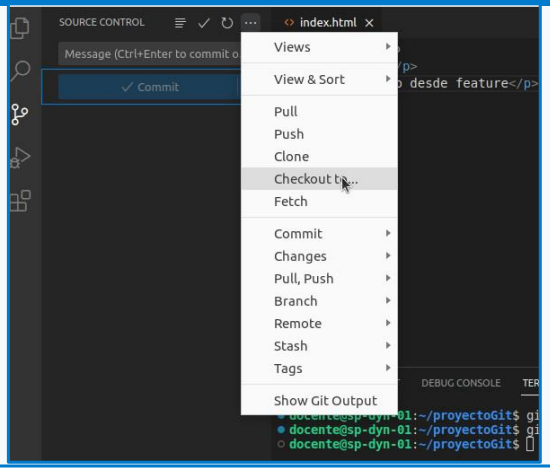
To share your local branch with the remote repository, click the 'Publish Branch' button in the Source Control view. This action automatically creates the corresponding branch on GitHub and uploads your committed changes.

```
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Equivalent Terminal Command:
  • git push origin feature1
```

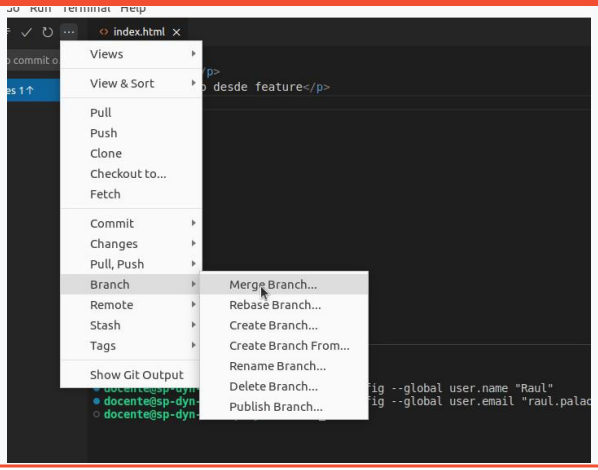


Merging Feature Branch into Develop

STEP 1: Switch to Develop Branch and pull



STEP 2: Merge Feature Branch

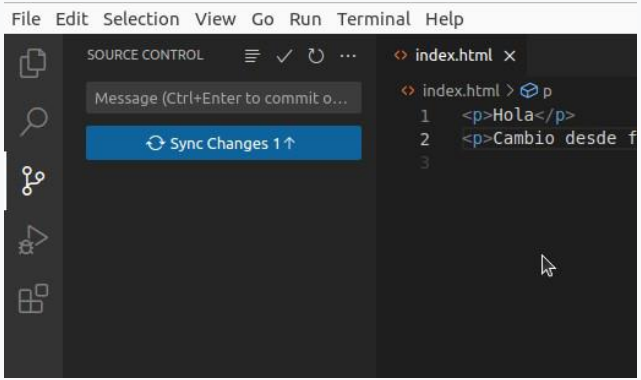


RESULT

- `git checkout develop`
- `git pull origin develop`
- `git merge feature1`

Updating the Remote Repository

After successfully merging the branches locally, you must update the remote repository. Click the 'Sync Changes' button in the Source Control view. This action pushes the merged master branch updates directly to GitHub.




```
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Equivalent Terminal Command:
```

- `git push origin develop`

Repeating the Process for Main / Master

After completing and merging your feature branch changes, simply repeat the same workflow for the master branch.

1. Switch back to your local master branch.

2. Sync and merge all latest updates.

3. Click 'Sync Changes' to push master directly to GitHub.



```
>_ Equivalent Terminal Commands:
```

- `git checkout master`
- `git pull origin master`
- `git merge develop`
- `git push origin master`